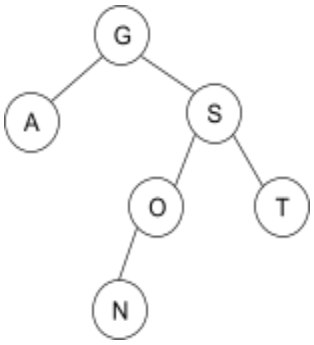
American Computer Science League

2024-2025 ● Contest 4: Binary Tree Arrays ● Senior Division

PROBLEM: The *Binary Search Tree* built from the letters **GASTON** is shown on the right. One way to represent this data structure is to use 3 arrays:

- an array of the *keys* in the order that they appear in the string
- an array of the position of each key's *left* child
- an array of the position of each key's *right* child



For each key, find where it must go in the tree and update the value of the parent's left or right array with the key's location. A value of -1 in the left or right array indicates that the node does not have that child. Here are the 3 arrays that are created from the letters **GASTON**:

	0	1	2	3	4	5	…
keys	**G**	**A**	**S**	**T**	**O**	**N**	
left	1	-1	4	-1	5	-1	
right	2	-1	3	-1	-1	-1	

In this program, you will be given a string to create a binary search tree from its letters. Once the tree is built using these arrays, find the following pieces of information for each tree: the number of nodes with only a left child, the number of nodes with only a right child, the number of leaf nodes, the internal path length, and the external path length. Output these 5 values as a string with each number separated by a single space.

EXAMPLE:

Input	`GASTON`
Output	`1 0 3 9 21`
Explanation	Initialize all elements in the left and right arrays to -1. The letter **G**: set key[0]='G'. The letter **A**: set key[1]='A'. A is the left child of G, so left[0]=1. The letter **S**: set key[2]='S'. S is the right child of G, so right[0]=2. The letter **T**: set key[3]='T'. T is the right child of S, so right[2]=3. The letter **O**: set key[4]='O'. O is the left child of S, so left[2]=4. The letter **N**: set key[5]='N'. N is the left child of O, so left[4]=5. The number of nodes with only a left child is 1 (letter **O**); the number of nodes with only a right child is 0; and the number of nodes that are leaves is 3 (**A**, **T**, **N**). The internal path length is the sum of the depths of each node, so the sum is 0 + 1 + 1 + 2 + 2 + 3 = 9. The external path length is the sum of the depths of

	the nodes that don't yet exist in the tree which is 2 + 2 (from node **A**) + 3 + 3 + 3 (from nodes **O** and **T**) + 4 + 4 (from node N) = 21. The output is `1 0 3 9 21`.

INPUT: Input one string that are all uppercase alphabetic characters and no more than 20 characters long.

OUTPUT: Output a string of numeric values, each separated by a single space, for each of the above 5 characteristics of the binary search tree.

SAMPLE INPUT:

```
1. GASTON
2. JASMINE
3. PETERPAN
4. RAPUNZEL
5. PINOCCHIO
6. MICKEYMOUSE
```

SAMPLE OUTPUT:

```
1. 1 0 3 9 21
2. 2 2 2 12 26
3. 3 0 3 14 30
4. 2 3 2 18 34
5. 2 2 3 22 40
6. 3 3 3 25 47
```

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TEST DATA

TEST INPUT:

```
1. CINDERELLA
2. MALEFICENT
3. DONALDDUCK
4. BUZZLIGHTYEAR
5. SLEEPINGBEAUTY
6. A
```

TEST OUTPUT:

```
1. 2 1 4 23 43
2. 1 4 3 27 47
3. 4 1 3 22 42
4. 4 0 5 37 63
5. 3 0 6 37 65
6. 0 0 1 0 2
```

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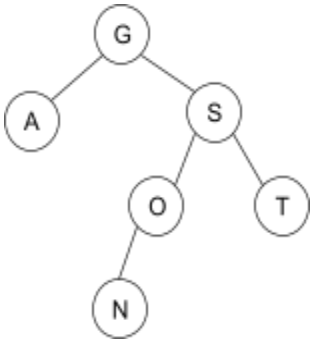
PROBLEM STATEMENT:



The *Binary Search Tree* built from the letters **GASTON** is shown on the right. One way to represent this data structure is to use 3 arrays:

- an array of the *keys* in the order that they appear in the string
- an array of the position of each key's *left* child
- an array of the position of each key's *right* child

For each key, find where it must go in the tree and update the value of the parent's left or right array with the key's location. A value of -1 in the left or right array indicates that the node does not have that child. Here are the 3 arrays that are created from the letters **GASTON**:

	0	1	2	3	4	5	…
keys	**G**	**A**	**S**	**T**	**O**	**N**	
left	1	-1	4	-1	5	-1	
right	2	-1	3	-1	-1	-1	

In this program, you will be given a string to create a binary search tree from its letters. Once the tree is built using these arrays, find the following pieces of information for each tree: the number of nodes with only a left child, the number of nodes with only a right child, the number of leaf nodes, the internal path length, and the external path length. Output these 5 values as a string with each number separated by a single space.

EXAMPLE:

Input:

```
GASTON
```

Output:

```
1 0 3 9 21
```

Explanation:

Initialize all elements in the left and right arrays to -1.

The letter **G**: set key[0]='G'.
The letter **A**: set key[1]='A'. A is the left child of G, so left[0]=1.
The letter **S**: set key[2]='S'. S is the right child of G, so right[0]=2.
The letter **T**: set key[3]='T'. T is the right child of S, so right[2]=3.
The letter **O**: set key[4]='O'. O is the left child of S, so left[2]=4.
The letter **N**: set key[5]='N'. N is the left child of O, so left[4]=5.

The number of nodes with only a left child is 1 (letter **O**); the number of nodes with only a right child is 0; and the number of nodes that are leaves is 3 (**A**, **T**, **N**). The internal path length is the sum of the depths of each node, so the sum is 0 + 1 + 1 + 2 + 2 + 3 = 9. The external path length is the sum of the depths of the nodes that don't yet exist in the tree which is 2 + 2 (from node **A**) + 3 + 3 + 3 (from nodes **O** and **T**) + 4 + 4 (from node **N**) = 21. The output is `1 0 3 9 21`.

TASK:

Complete the function **analyzeTree**

- The function has 1 parameter: a string, *word*, that is used to build the binary search tree
- The function returns a string of 5 numeric values which are identified above, each separated by a single space

You may create additional functions that are called from **analyzeTree** if needed in solving the problem.

CONSTRAINTS:

The word will be all uppercase letters and no more than 20 characters in length.

DATA PROVIDED:

There are 6 sets of Sample Data for debugging and 6 sets of Test Data for scoring. The test cases will vary in difficulty. You should create sample data of your own to fully test your program.

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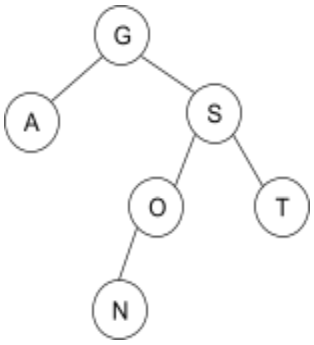
问题：由字符串 GASTON 构建的*二叉搜索树*如右图所示。该数据结构可通过以下三个数组表示：

- 一个按字符串中出现顺序存储*键*的数组
- 一个数组记录每个键*左*子节点的位置
- 一个数组记录每个键*右*子节点的位置



插入新键时，需确定其在树中的位置，并将其父节点左子数组或右子数组的值更新为新键的位置。若左子数组或右子数组中的值为 -1，则表示该节点对应的子节点不存在。

以下是由字符串 GASTON 生成的三个数组：

	0	1	2	3	4	5	…
键	**G**	**A**	**S**	**T**	**O**	**N**	
左子节点	1	-1	4	-1	5	-1	
右子节点	2	-1	3	-1	-1	-1	

本程序将提供一个字符串，你需要根据其字母来构建二叉搜索树。树构建完成后，请为每棵树找出以下信息：仅含左子节点的节点数、仅含右子节点的节点数、叶节点数、内部路径长度和外部路径长度。将这 5 个数值以字符串形式输出，各数值之间用单个空格分隔。

示例：

输入	GASTON
输出	1 0 3 9 21
解析	将左子节点数组和右子节点数组的所有元素初始化为 -1。 字母 **G**：设置 key[0]='G'。 字母 **A**：设置 key[1]='A'。A 是 G 的左子节点，故 left[0]=1。 字母 **S**：设置 key[2]='S'。S 是 G 的右子节点，故 right[0]=2。 字母 **T**：设置 key[3]='T'。T 是 S 的右子节点，故 right[2]=3。 字母 **O**：设置 key[4]='O'。O 是 S 的左子节点，故 left[2]=4。 字母 **N**：设置 key[5]='N'。N 是 O 的左子节点，故 left[4]=5。 仅含左子节点的节点数为 1（字母 **O**）；仅含右子节点的节点数为 0；叶节点数为 3 (**A**, **T**, **N**）。内部路径长度为各节点深度之和，计算结果为 0 + 1 + 1 + 2 + 2 + 3 = 9。

	外部路径长度为树中不存在节点的深度之和，计算过程为：2 + 2（来自节点 **A**） + 3 + 3 + 3（来自节点 **O** 和节点 **T**） + 4 + 4（来自节点 **N**）= 21。 输出结果为 `1 0 3 9 21`。

输入：输入一个全由大写字母组成的字符串，长度不超过 20 个字符。

输出：输出一串数值，每个数值之间用单个空格分隔，分别表示上述二叉搜索树的 5 个特征。

样例输入：

```
1. GASTON
2. JASMINE
3. PETERPAN
4. RAPUNZEL
5. PINOCCHIO
6. MICKEYMOUSE
```

样例输出：

```
1. 1 0 3 9 21
2. 2 2 2 12 26
3. 3 0 3 14 30
4. 2 3 2 18 34
5. 2 2 3 22 40
6. 3 3 3 25 47
```

测试数据

测试输入：

```
1. CINDERELLA
2. MALEFICENT
3. DONALDDUCK
4. BUZZLIGHTYEAR
5. SLEEPINGBEAUTY
6. A
```

测试输出：

```
1. 2 1 4 23 43
2. 1 4 3 27 47
3. 4 1 3 22 42
4. 4 0 5 37 63
5. 3 0 6 37 65
6. 0 0 1 0 2
```

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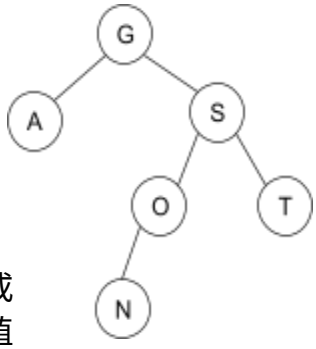
问题陈述:

由字符串 GASTON 构建的*二叉搜索树*如右图所示。该数据结构可通过以下三个数组表示：



- 一个按字符串中出现顺序存储*键*的数组
- 一个数组记录每个键*左*子节点的位置
- 一个数组记录每个键*右*子节点的位置

插入新键时，需确定其在树中的位置，并将其父节点左子数组或右子数组的值更新为新键的位置。若左子数组或右子数组中的值为 -1，则表示该节点对应子节点不存在。

以下是由字符串 GASTON 生成的三个数组：

	0	1	2	3	4	5	…
键	**G**	**A**	**S**	**T**	**O**	**N**	
左子节点	1	-1	4	-1	5	-1	
右子节点	2	-1	3	-1	-1	-1	

本程序将提供一个字符串，你需要根据其字母来构建二叉搜索树。树构建完成后，请为每棵树找出以下信息：仅含左子节点的节点数、仅含右子节点的节点数、叶节点数、内部路径长度和外部路径长度。将这 5 个数值以字符串形式输出，各数之间用单个空格分隔。

示例:

输入：

```
GASTON
```

输出：

```
1 0 3 9 21
```

解析：

将左子节点数组和右子节点数组的所有元素初始化为 -1。

字母 **G**：设置 key[0]=’G’。
字母 **A**：设置 key[1]=’A’。A 是 G 的左子节点，故 left[0]=1。
字母 **S**： 设置 key[2]=’S’。S 是 G 的右子节点，故 right[0]=2。
字母 **T**：设置 key[3]=’T’。T 是 S 的右子节点，故 right[2]=3。
字母 **O**：设置 key[4]=’O’。O 是 S 的左子节点，故 left[2]=4。
字母 **N**：设置 key[5]=’N’。N 是 O 的左子节点，故 left[4]=5。

仅含左子节点的节点数为 1（字母 **O**）；仅含右子节点的节点数为 0 ；叶节点数为 3 (**A**, **T**, **N**）。内部路径长度为各节点深度之和，计算结果为 0 + 1 + 1 + 2 + 2 + 3 = 9。外部路径长度为树中不存在节点的深度之和，计算过程为：2 + 2（来自节点 **A**） + 3 + 3 + 3（来自节点 **O** 和节点 **T** ） + 4 + 4（来自节点 **N** ） = 21。
输出结果为 `1 0 3 9 21`。

任务:
完成函数 **analyzeTree**

- 该函数有 1 个参数：一个字符串，*word*，用于构建二叉搜索树。
- 该函数返回一个字符串，包含 5 个用单个空格分隔的数值，这些数值如上文所述。

如有需要，你可以构建额外的函数供 **analyzeTree** 调用来解答此问题。

约束条件：

该字符串将全部由大写字母组成，且长度不超过 20 个字符。

数据说明:
包含 6 组用于调试的样例数据和 6 组用于评分的测试数据。测试案例难度各不相同。您应该自行创建样例数据以充分验证程序。